0121. Best Time to Buy and Sell Stock

Easy · Sliding Window

leetcode.com/problems/best-time-to-buy-and-sell-stock

Problem

- Prices[i] is the price on day i. Buy once and sell once later; return the max profit (0 if none).
- Example: [7,1,5,3,6,4] → 5 (buy at 1, sell at 6).

Intuition

- For each selling day, the best buy is the minimum price seen so far.
- Track that running minimum (the window's left edge) and the best difference.

Brute Force

- All pairs i < j: O(n^2).

Optimal Approach

- MinPrice = +infinity, best = 0.
- For each p: minPrice = min(minPrice, p); best = max(best, p - minPrice).

Worked Example

```
prices = [7,1,5,3,6,4].

p=7 min=7 best=0
p=1 min=1 best=0
p=5 min=1 best=4
p=3 min=1 best=4
p=6 min=1 best=5
p=4 min=1 best=5
```

Java Solution

```java
public class BestTimeToBuyAndSellStock {
    public int maxProfit(int[] prices) {
        int min = Integer.MAX_VALUE, best = 0;
        for (int p : prices) {
            min = Math.min(min, p);
            best = Math.max(best, p - min);
        }
        return best;
    }
}
```

Complexity

- Time: O(n). Space: O(1).

Pitfalls

- You cannot sell before buying, so a later minimum must not pair with an earlier price.